

Introduction to Microeconomics

Lecture 8—Consumer Theory: Preferences

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

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A model of consumer Preferences

- We start with a set of objects (bundles) X . Elements from from this set, like x or y , are the choices that the consumer can make.
- Most applications consider $X \subset \mathbb{R}_+^n$.
- In this class, $X = \mathbb{R}_+^2$. A bundle x is then $x = (x_1, x_2)$. Interpret it as: x_1 is the quantity of good 1 and x_2 is the quantity of good 2.

Consumer preferences: The Questionnaire Q

Consists of a long list of questions of the following form.

$Q(x, y)$, x and y are different bundles. How do you compare x to y ? Check one and only one of the following three options:

- ☐ I strictly prefer x to y ($x \succ y$).
- ☐ I strictly prefer y to x ($y \succ x$).
- ☐ I am indifferent between x and y ($x \sim y$).

A model of consumer preferences is just a long list of questions of this type and their answers!

What is a legal answer to the questionnaire?

- We need to put some restrictions (assumptions) on answers.

Restriction 1: Completeness

For any question $Q(x, y)$ exactly one of the boxes is checked. We don't allow

- Refraining to answer (no box checked).
- Checking more than one box.
- This means we exclude attitudes like:
 - They are not comparable.
 - I don't know what x is.
 - I have no opinion.
 - I prefer x over y and y over x .
- We are also not allowing attitudes like:
 - It depends on what others think.
 - It depends on the circumstances (frame).
 - Any consideration that includes **intensity**, like I somewhat prefer x , or I love x and hate y .

Restriction 2: Transitivity

Suppose I have bundles x, y, z such that $Q(x, y) : x \succ y$, and $Q(y, z) : y \succ z$. Then this implies that your answer to $Q(x, z)$ should be $x \succ z$.

Similarly, if $Q(x, y)$ is $x \sim y$ and $Q(y, z)$ is $y \sim z$ then in $Q(x, z)$ your answer should be $x \sim z$.

- Questionnaire Q could have been rephrased instead with the following type of questions:
 - $Q(x, y)$, x and y are different bundles. How do you compare x to y ? Check one and only one of the following three options:
 - ☐ I consider x at least as good as y ($x \succeq y$).
 - ☐ I consider y at least as good as x ($y \succeq x$).
- What is the relationship between the answers to questionnaire Q and the one above?
 - If $x \succeq y$ and $y \succeq x$ then $x \sim y$.
 - If $x \succeq y$ but $y \not\succeq x$ then $x \succ y$.
 - If $y \succeq x$ but $x \not\succeq y$ then $y \succ x$.

Suppose the set of bundles $X \subset \mathbb{R}_+^n$ consists of *goods that are desirable*.

Restriction 3: Monotonicity or “more-is-better”

If $x \geq y$, where $\geq$ denotes the component-wise standard order, then $x \succsim y$.

If $x > y$, where $>$ denotes the component-wise strict order, then $x \succ y$.

Example: say good 1 is food and good 2 is clothing. $x = (30, 40)$ and $y = (20, 30)$. If preferences satisfy monotonicity, then the answer to $Q(x, y)$ should be $x \succ y$.

- Any relation R over X is called a **preference relation** if it consists of a set of legal answers to the questionnaire Q , that is, if it satisfies completeness and transitivity.

- For a bundle x the **indifference curve at x** is the set of all bundles that are indifferent to x :

$$IC(x) = \{y \in X : y \sim x\}$$

- The strictly preferred set to x is the set of all bundles that are strictly preferred to x :

$$\succ(x) = \{y \in X : y \succ x\}$$

- The strictly worse set to x is the set of all bundles that are worse to x :

$$\prec(x) = \{y \in X : x \succ y\}$$

- Completeness implies that the indifference curve map is **full**, through each bundle there is an indifference curve passing through it.
- Transitivity implies that the indifference curves **cannot intersect!**
- Monotonicity implies that the indifference curves **are not thick!**

- Although preference relations provide a model of the consumers preferences, they are somewhat cumbersome to work with.
- Recall: a preference relation provides an **ordering** over the consumption bundles.
- Idea: can we get some mathematical object that is more tractable to work with, while still preserving the idea that a preference relation provides an ordering over consumption bundles?
- The answer is **Yes**!
- We say that a **utility function** $u : X \rightarrow \mathbb{R}$ represents the preference relation $\succsim$ over X if for every $x, y \in X$, $x \succsim y \Leftrightarrow u(x) \geq u(y)$.
- Given a preference relation, can we always ensure that a utility function exists?
 - If $\succsim$ is complete and transitive and X is finite or (countable), the answer is Yes.
 - On the other hand, if X is infinite, like $\mathbb{R}_+^2$, then we need an extra technical restriction: **continuity**, but this will be discussed in higher level courses.

Utility is an ordinal concept, not a cardinal one.

- Suppose that a preference relation $\succsim$ can be represented by the utility function u .
- Do the numbers $u(x)$ for each $x \in X$ matter?
- The answer is no, what matters is the order (Ranking), not the number!
- Utility is ordinal, not cardinal (the magnitude matters).
- The previous idea can be formally stated with the concept of a **monotonic transformation**:
 - Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing function, i.e. if $a \geq b$ then $f(a) \geq f(b)$.
 - Let u be a utility function representing $\succsim$. Then $v(x) = f(u(x))$ represents the same preferences as u .
 - $x \succsim y \Leftrightarrow u(x) \geq u(y) \Leftrightarrow v(x) \geq v(y)$.

- With the help of a utility function, it becomes easy to define an indifference curve:

$$x \sim y \Leftrightarrow u(x) = u(y)$$

- Then the indifference curve passing through x is the set of consumption bundles that achieve the same level of utility $u(x)$:

$$IC(x) = \{y \in X : u(y) = u(x)\}$$

- We can define in a similar way the strictly preferred and the strictly worse sets.

The Marginal Rate of Substitution

- It is the absolute value of the slope of the indifference curve at a particular point.
- It measures the cost of opportunity of good 1 in terms of good 2 while staying on the same indifference curve.
- Intuitively, it is the personal opportunity cost (in terms of the consumer's preferences) of remaining in the same indifference curve, measured as units of good 2 that must be given up to get one more unit of good 1.
- How? Let's compute the total change in utility, when the utility function is $u(x_1, x_2)$:

$$\Delta U = \frac{\Delta U}{\Delta x_1} \Delta x_1 + \frac{\Delta U}{\Delta x_2} \Delta x_2$$

- At an indifference curve, utility is not changing, so $\Delta U = 0$, this leads to:

$$\frac{\Delta x_2}{\Delta x_1} = - \frac{\frac{\Delta U}{\Delta x_1}}{\frac{\Delta U}{\Delta x_2}}$$

- The **Marginal Rate of Substitution** is the absolute value of the quantity above.

Convexity

Preferences that are convex exhibit a preference for more balanced consumption bundles.